

Youngjoo Shin

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Summary

Youngjoo Shin is an Associate Professor in the School of Cybersecurity at Korea University, where he directs the Computer Systems Security Lab. His research investigates the security of modern computer systems — from CPU microarchitecture to operating systems, virtualization platforms, and cloud infrastructure — with a particular focus on side-channel and transient-execution attacks, OS kernel defenses, and practical privacy-preserving system design. Before joining Korea University in 2020, he was an Assistant Professor at Kwangwoon University (2017–2020), a Senior Researcher at the National Security Research Institute (2008–2017), and a Researcher at LG Electronics' Digital Media Laboratory (2007–2008).

Interests

System Security, Microarchitectural Side-channel Attacks, Transient Execution Attacks, OS & Hypervisor Security, Cloud & Container Security, Network Security, Applied Cryptography

Education

KAIST <i>Ph.D in Computer Science</i>	<i>Korea</i>
KAIST <i>M.S in Computer Science</i>	<i>Korea</i>
Korea University <i>B.S in Computer Science and Engineering</i>	<i>Korea</i>

Experience

Associate Professor <i>School of Cybersecurity, Korea University</i>	<i>2022.3 – present</i>
Assistant Professor <i>School of Cybersecurity, Korea University</i>	<i>2020.9 – 2022.2</i>
Assistant Professor <i>Kwangwoon University</i>	<i>2017.3 – 2020.8</i>
Senior Researcher <i>National Security Research Institute (NSR)</i>	<i>2008.4 – 2017.2</i>
Researcher <i>Digital Media Laboratory, LG Electronics</i>	<i>2007.11 – 2008.3</i>
Research Assistant <i>Mobile Multimedia Platform Center, KAIST</i>	<i>2006.3 – 2007.12</i>

Publications

- Input-Agnostic Runtime Metric Based Memory Optimization for Serverless Functions** Jan 2026
Seunghun Kim, Youngjoo Shin
(IEEE International Conference on Cloud Computing (CLOUD))
- TIMESLICE-SANDWICH: A GPU Side-channel Attack Exploiting Time-Sliced Scheduling** Jan 2026
Hodong Kim, Gyeongsup Lim, Seunghee Shin, Youngjoo Shin, Junbeom Hur
(USENIX Security Symposium)
- SysDiver: Lightweight and Fast Static Analysis for Windows Kernel Drivers** Jan 2026
Chanhee Park, Dongjoo Kim, Youngjoo Shin
(ACM Asia Conference on Computer and Communications Security (ASIACCS))
- Empirical Study on BMC Firmware Vulnerabilities: Root Causes and Architectural Insights** Jan 2026
Jihye Lee, Chanhee Park, Youngjoo Shin
(International Conference on Information Networking (ICOIN))
- PathFault: Automated Exploit Generator for Web Services via HTTP Message Parser Discrepancies** Jan 2025
Juryeok Kim, Youngjoo Shin
(International Conference on Information Security and Cryptology (ICISC))
- MimicCall: Bypassing System Call Filters via Kernel Function Redundancy** Jan 2025
Songah Joo, Minchan Park, Hyerean Jang, Youngjoo Shin
(Annual Computer Security Applications Conference (ACSAC))
- A Survey of Speculative Load Prediction Attacks Exploiting Memory Disambiguation Units** Jan 2025
Selynn Kim, Taehun Kim, Youngjoo Shin
(World Conference on Information Security Applications (WISA))
- PodBeater: Exploiting Multi-Value Affinity for Efficient Co-Location Attacks in Kubernetes** Jan 2025
Yiju Jung, Hyerean Jang, Youngjoo Shin
(World Conference on Information Security Applications (WISA))
- Maximizing GPU Parallelism for a High-performance Cryptanalysis System** Jan 2025
Sangyub Kim, Youngjoo Shin
(International Conference on Information Networking (ICOIN))
- T-Time: A Fine-grained Timing-based Controlled-Channel Attack against Intel TDX** Jan 2025
Woomin Lee, Taehun Kim, Seunghee Shin, Junbeom Hur, Youngjoo Shin
(European Symposium on Research in Computer Security (ESORICS))
- Cache Demote for Fast Eviction Set Construction and Page Table Attribute Leakage** Jan 2025
Taehun Kim, Hyerean Jang, Youngjoo Shin
(European Symposium on Research in Computer Security (ESORICS))

Vulnerable Intel GPU Context: Prohibit Complete Context Restore by Modifying Kernel Driver Wonseok Choi, Youngjoo Shin (ACM Asia Conference on Computer and Communications Security (ASIACCS))	Jan 2025
A Survey of Side-Channel Attacks on Branch Prediction Units Jihoon Kim, Hyerean Jang, Youngjoo Shin (ACM Computing Surveys)	Jan 2025
BranchCloak: Mitigating Side-Channel Attacks on Directional Branch Predictors Jihoon Kim, Hyerean Jang, Youngjoo Shin (Electronics)	Jan 2025
FuzzyBin: Enhanced Border Binary Identification by Leveraging Fuzzy Hashing Algorithms Dongsoo Kim, Hyerean Jang, Youngjoo Shin (IEEE Access)	Jan 2025
SysBumps: Exploiting Speculative Execution in System Calls for Breaking KASLR in macOS for Apple Silicon Hyerean Jang, Taehun Kim, Youngjoo Shin (ACM Conference on Computer and Communications Security (CCS))	Jan 2024
POSTER: Beyond the Filter: Advanced Mimicry Attack to Bypass System Call Filtering Songah Joo, Hyerean Jang, Youngjoo Shin (Annual Computer Security Applications Conference (ACSAC))	Jan 2024
HBRA: A History-Based Remote Attestation Approach to Resolve Malicious Verifiers in IoT Euiseong Moon, Youngjoo Shin (International Conference on ICT Convergence (ICTC))	Jan 2024
POSTER: On the Feasibility of Inferring SGX Execution through PMU Woomin Lee, Taehun Kim, Youngjoo Shin (ACM Asia Conference on Computer and Communications Security (ASIACCS))	Jan 2024
S-ZAC: Hardening Access Control of Service Mesh using Intel SGX for Zero Trust in Cloud Changhee Han, Taehun Kim, Woomin Lee, Youngjoo Shin (Electronics)	Jan 2024
Exploiting Memory Page Management in KSM for Remote Memory Deduplication Attack Seungyeon Bae, Taehun Kim, Woomin Lee, Youngjoo Shin (World Conference on Information Security Applications (WISA))	Jan 2023
DevIOUs: Device-Driven Side-Channel Attacks on the IOMMU Taehun Kim, Hyeongjin Park, Seokmin Lee, Seunghee Shin, Junbeom Hur, Youngjoo Shin (IEEE Symposium on Security and Privacy (S\&P))	Jan 2023

MicroCFI: Microarchitecture-level Control-Flow Restrictions for Spectre Mitigation Hyerean Jang, Youngjoo Shin (IEEE Access)	Jan 2023
Deep Learning based Detection for Multiple Cache Side-channel Attacks Hodong Kim, Changhee Hahn, Hyunwoo J. Kim, Youngjoo Shin, Junbeom Hur (IEEE Transactions on Information Forensics and Security)	Jan 2023
GAN-based patient information hiding for an ECG authentication system Youngshin Kang, Geunbo Yang, Heesang Eom, Seungwoo Han, Suwhan Baek, Seungil Noh, Youngjoo Shin, Cheolsoo Park (Biomedical Engineering Letters)	Jan 2023
Fuzzing of Embedded Systems: A Survey Joobeom Yun, Fayozbek Rustmov, Juhwan Kim, Youngjoo Shin (ACM Computing Surveys)	Jan 2023
A Performance Evaluation of IPsec with Post-Quantum Cryptography Seungyeon Bae, Yuseong Chang, Hyeongjin Park, Youngjoo Shin (International Conference on Information Security and Cryptology (ICISC))	Jan 2022
Avengers, Assemble! Survey of WebAssembly Security Solutions Minseo Kim, Hyerean Jang, Youngjoo Shin (IEEE International Conference on Cloud Computing (CLOUD))	Jan 2022
Broken Heart: Privacy Leakage Analysis on ECG-based Authentication Schemes Seungil Noh, Jaehan Kim, Seokmin Lee, Youngshin Kang, Cheolsoo Park, Youngjoo Shin (Security and Communication Networks)	Jan 2022
ThermalBleed: A Practical Thermal Side-Channel Attack Taehun Kim, Youngjoo Shin (IEEE Access)	Jan 2022
Quantitative Analysis on Attack Capacity in Meltdown-type Attacks Seokmin Lee, Taehun Kim, Youngjoo Shin (World Conference on Information Security Applications (WISA))	Jan 2021
A Secure and Privacy-Preserving ECG-based Personal Authentication Youngshin Kang, Hyeonbin Lee, Youngjoo Shin, Cheolsoo Park (IEEE International Conference on Smart Internet of Things (SmartIoT))	Jan 2021
(Extended Abstract) DCUIP Poisoning Attack in Intel x86 Processors Youngjoo Shin (International Conference on Information Networking (ICOIN))	Jan 2021
Constructing Covert Channel on Intel CPU-iGPU Platform Taehun Kim, Taehyun Kim, Youngjoo Shin (International Conference on Information Networking (ICOIN))	Jan 2021
Breaking KASLR Using Memory Deduplication in Virtualized Environments Taehun Kim, Jaehan Kim, Youngjoo Shin (Electronics)	Jan 2021

Runtime Randomized Relocation of Crypto Libraries for Mitigating Cache Attacks Youngjoo Shin, Joobeom Yun (IEEE Access)	Jan 2021
DCUIP Poisoning Attack in Intel x86 Processors Youngjoo Shin (IEICE Transactions on Information and Systems)	Jan 2021
Multibyte microarchitectural data sampling and its application to session key extraction attacks Youngjoo Shin (IEEE Access)	Jan 2021
Return of Version Downgrade Attack in the Era of TLS 1.3 Sangtae Lee, Youngjoo Shin, Junbeom Hur (ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT))	Jan 2020
Software-based Random Number Generators for Resource Constrained Devices Seungil Noh, Youngjoo Shin (International Conference on ICT Convergence (ICTC))	Jan 2020
(Poster) Mitigating Memory Sharing-based Side-channel Attack by Embedding Random Values in Binary for Cloud Environment Taehun Kim, Youngjoo Shin (ACM Asia Conference on Computer and Communications Security (ASIACCS))	Jan 2020
GPU Side-channel Attacks are Everywhere: A Survey Taehun Kim, Youngjoo Shin (IEEE International Conference on Consumer Electronics - Asia (ICCE-Asia))	Jan 2020
Inferring Firewall Rules by Cache Side-channel Analysis in Network Function Virtualization Youngjoo Shin, Dongyoung Koo, Junbeom Hur (IEEE International Conference on Computer Communications (INFOCOM))	Jan 2020
Cache Side-Channel Attack on Mail User Agent Hodong Kim, Hyundo Yoon, Youngjoo Shin, Junbeom Hur (International Conference on Information Networking (ICOIN))	Jan 2020
Decentralized Server-aided Encryption for Secure Deduplication in Cloud Storage Youngjoo Shin, Dongyoung Koo, Joobeom Yun, Junbeom Hur (IEEE Transactions on Services Computing)	Jan 2020
Toward Serverless and Efficient Encrypted Deduplication in Mobile Cloud Computing Environments Youngjoo Shin, Junbeom Hur, Dongyoung Koo, Joobeom Yun (Security and Communication Networks)	Jan 2020
Lightweight and Seamless Memory Randomization for Mission-Critical Services in Cloud Platform Joobeom Yun, Ki-Woong Park, Dongyoung Koo, Youngjoo Shin (Energies)	Jan 2020

Real-Time Detection for Cache Side Channel Attack using Performance Counter Monitor Jonghyeon Cho, Taehun Kim, Soojin Kim, Miok Im, Taehyun Kim, Youngjoo Shin (Applied Sciences)	Jan 2020
FPGA reverse engineering in Vivado design suite based on X-ray project Hoyoung Yu, Hyung-Min Lee, Youngjoo Shin, Youngmin Kim (International SoC Design Conference (ISOCC))	Jan 2019
Real-Time Detection on Cache Side Channel Attacks using Performance Counter Monitor JongHyeon Cho, TaeHyun Kim, Youngjoo Shin (International Conference on ICT Convergence (ICTC))	Jan 2019
High efficiency, low-noise Meltdown attack by using a return stack buffer TaeHyun Kim, Youngjoo Shin (ACM Asia Conference on Computer and Communications Security (ASIACCS))	Jan 2019
Poster: FLUSH+REALOD Cache Side-Channel Attack on Mail User Agent Hodong Kim, Hyundo Yoon, Youngjoo Shin, Junbeom Hur (Network and Distributed System Security Symposium (NDSS))	Jan 2019
Reinforcing Meltdown Attack By using Return Stack Buffer TaeHyun Kim, Youngjoo Shin (IEEE Access)	Jan 2019
A VM-Based Detection Framework against Remote Code Execution Attacks for Closed Source Network Devices Youngjoo Shin (Applied Sciences)	Jan 2019
Cross-VM Cache Timing Attacks on Virtualized Network Functions Youngjoo Shin (IEICE Transactions on Information and Systems)	Jan 2019
Unveiling Hardware-based Data Prefetcher, a Hidden Source of Information Leakage Youngjoo Shin, Hyung Chan Kim, Dokeun Kwon, Ji Hoon Jeong, Junbeom Hur (ACM Conference on Computer and Communications Security (CCS))	Jan 2018
Privacy-preserving and Updatable Block-level Data Deduplication in Cloud Storage Services Hyungjune Shin, Dongyoung Koo, Youngjoo Shin, Junbeom Hur (IEEE International Conference on Cloud Computing (CLOUD))	Jan 2018
Improving Security and Reliability in Merkle Tree-based Online Data Authentication with Leakage Resilience Dongyoung Koo, Youngjoo Shin, Joobeom Yun, Junbeom Hur (Applied Sciences)	Jan 2018
Fast and Secure Implementation of Modular Exponentiation for Mitigating Fine-grained Cache attacks Youngjoo Shin (Applied Sciences)	Jan 2018

- An Online Data-Oriented Authentication based on Merkle Tree with Improved Reliability** Jan 2017
Dongyoung Koo, Youngjoo Shin, Joobeom Yun, Junbeom Hur
(IEEE International Conference on Web Services (ICWS))
- Secure Data Deduplication with Dynamic Ownership Management in Cloud Storage (Extended Abstract)** Jan 2017
Junbeom Hur, Dongyoung Koo, Youngjoo Shin, Kyungtae Kang
(IEEE International Conference on Data Engineering (ICDE))
- Practical Data Outsourcing Framework with Provably Secure Deduplication in Untrusted Remote Storage** Jan 2017
Dongyoung Koo, Youngjoo Shin, Junbeom Hur
(International Conference on Platform Technology and Service (PlatCon))
- A Survey of Secure Data Deduplication Schemes for Cloud Storage Systems** Jan 2017
Youngjoo Shin, Dongyoung Koo, Junbeom Hur
(ACM Computing Surveys)
- Decentralized Server-aided Encryption for Secure Deduplication in Cloud Storage** Jan 2017
Youngjoo Shin, Dongyoung Koo, Joobeom Yun, Junbeom Hur
(IEEE Transactions on Services Computing)
- Secure Proof of Storage with Deduplication for Cloud Storage Systems** Jan 2017
Youngjoo Shin, Dongyoung Koo, Junbeom Hur, Joobeom Yun
(Multimedia Tools and Applications)
- CLDSafe: An Efficient File Backup System in Cloud Storage against Ransomware** Jan 2017
Joobeom Yun, Junbeom Hur, Youngjoo Shin, Dongyoung Koo
(IEICE Transactions on Information and Systems)
- Privacy-Preserving Aggregation and Authentication of Multi-source Smart Meters in a Smart Grid System** Jan 2017
Dongyoung Koo, Youngjoo Shin, Junbeom Hur
(Applied Sciences)
- An efficient stream cipher for resistive RAM** Jan 2017
Joobeom Yun, Ki-Woong Park, Youngjoo Shin, Hee-Dong Kim
(IEICE Electronics Express)

invited talks

2026: NetSec-KR 2026, IITP Special Session (invited talk)
2025: WDSC 2025 (invited lecture); Drone Cybersecurity Seminar; Unification & Sharing Policy Research Seminar
2024: NetSec-KR 2024
2023: KAIST Colloquium; KSC 2023; ROK Navy Cybersecurity Seminar; WebAssembly Security Seminar
2022: Hansung University
2021: NSR; Hansung University; Sejong University
2020: Side-Channel Analysis Workshop; NSR; Hoseo University; SeoulTech; Hansung University; Sejong University; Kookmin University
2019: KAIST; Korea University; SeoulTech; Sejong University; Hansung University; Kookmin University; WISC 2019; Side-Channel Analysis Workshop
2018: NetSec-KR 2018; KICS; ETRI; NSR; WISC 2018; Side-Channel Analysis Workshop; Sejong / Kookmin / Hansung / Hannam Universities
2017: KCC 2017; SDN/NFV Forum Security WG; NSR; Korea / Kookmin / Hansung Universities

government & public-sector advisory

- IITP Preliminary Feasibility Study, Intelligent Cyber-Shield Dome Program — Subcommittee Member (2025–)
- Cloud Security Assurance Program (CSAP) Certification Committee (2023–2024)
- National Security Research Institute (NSR) — Digital-Signature Advisory (2022)
- National Intelligence Service — Mid-/Long-Term Security Strategy Advisory (2019)
- IITP Preliminary Feasibility Study on Connected-Device Security — Committee Member (2017)

peer review & evaluation

- Korea Data Agency (KoData) — KOSDAQ Technology Assessment Advisor, ICT Sector
- Korea Internet & Security Agency (KISA) — Evaluation Committee Member (2017, 2018)

program committee

- 2026: PST 2026; SECRIPT 2026; SecureComm 2026 (EAI); IFIP SEC 2026
- 2024: ICISC 2024; SmartSP 2024
- 2023: ICISC 2023
- 2022: WoNEXT 2022; CISC-W 2022
- 2020: KCS 2020
- 2018: Security 2018 (사이버보안 논문 공모전)

standards, guidelines & industry

- Korea Association for Industrial Technology Security (KAITS) — Contribution/Advisory on The Future of Sovereign Cloud
- KISA — Key-Management Guidelines Review Advisory
- IDEC — External Advisor (2024)
- LG U+ — Security Advisory (2019)
- KANI; SDN/NFV Forum Security Working Group (2017)

Academia

- Kwangwoon University — Faculty Recruitment Review Committee (2024)
- Kookmin University — Faculty Review Committee (2018)